

IDEABATIC

SMILE GO Project

Uniform Cooling Solutions

Interim Presentation by Karen, Kerry, and Kavita



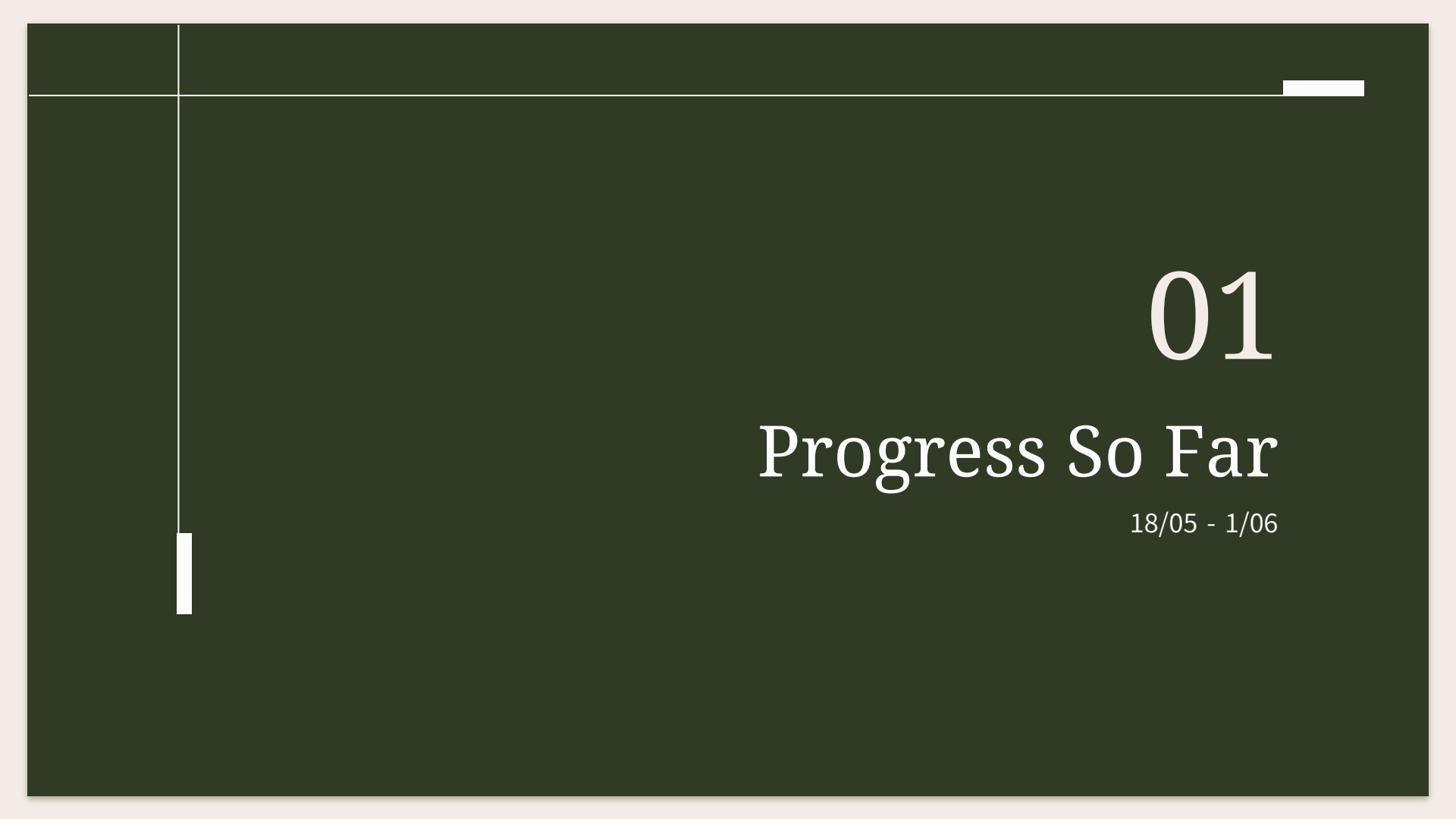
Proposal Recap

SMILE GO - Investigating the causes of Asymmetric Cooling & potential ways to reduce it

- In thermal tests the SMILE coolers have shown significant temp difference between top and bottom vaccine carousel chambers
- Issue because vaccines must be stored in small range (2-8°C)

What we will do:

- Design & test cooler inserts
- Investigate impact of convection
- Investigate any other potential causes
- Consider new designs based on our findings

A dark green background with a white vertical line on the left and a white horizontal line near the top. There are also two small white rectangular bars, one at the top right and one at the bottom left.

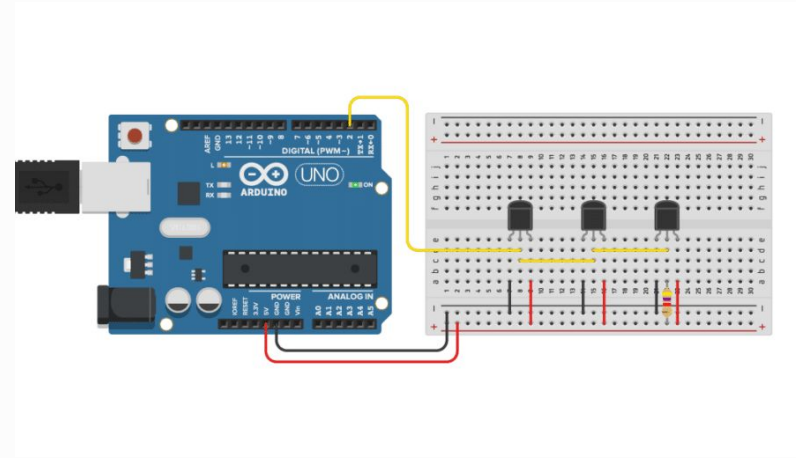
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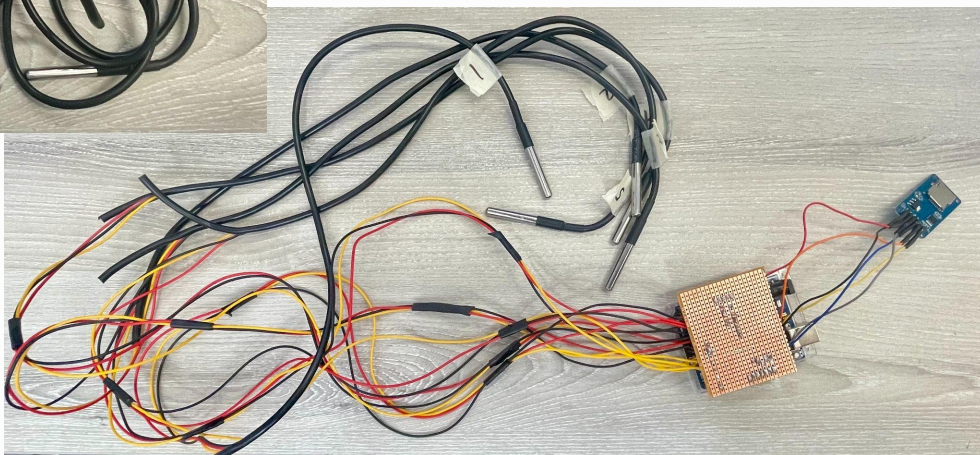
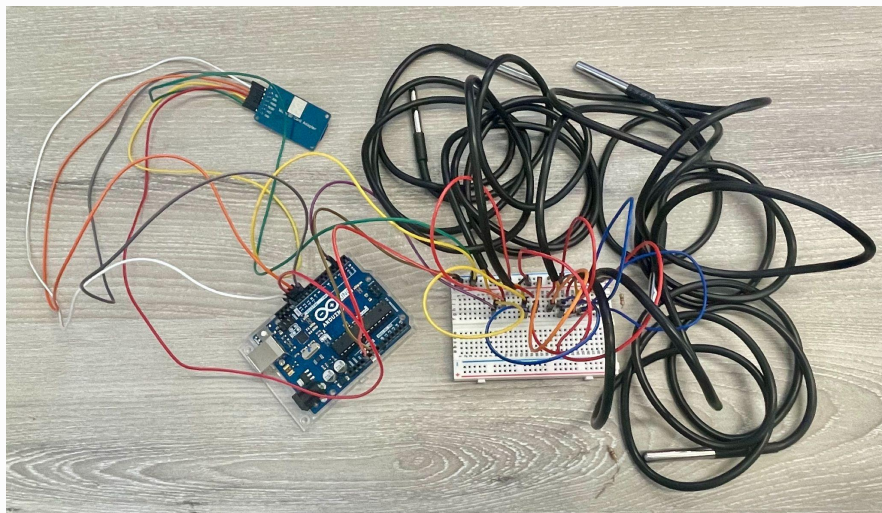
Progress So Far

18/05 - 1/06

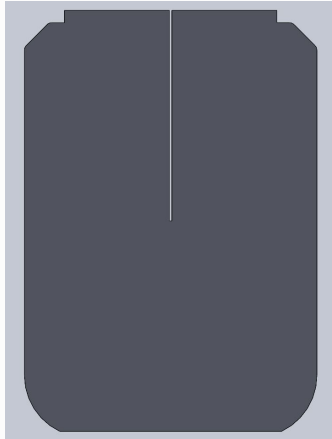
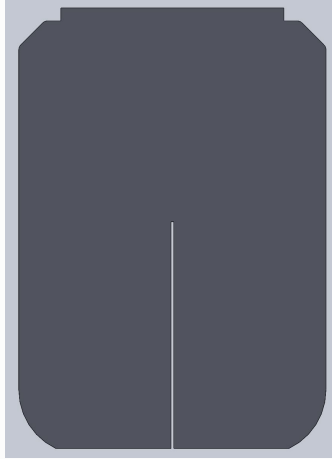
Electronics

Supplies	Quantity	Source
DS18B20	x5 (6)	RS components
breadboard	x1	CUED
Arduino Uno	x1	CUED
SD card module	x1	CUED
4.7k resistor	x1	CUED
RTC	x1	RS components

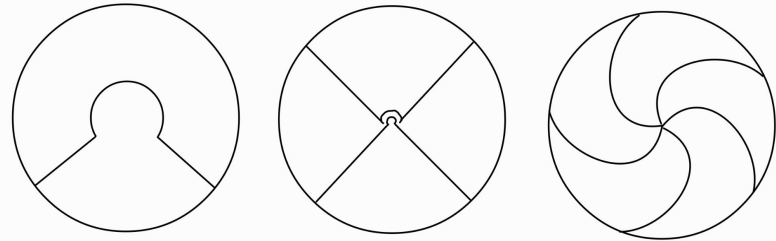




Insert Designs



- Multiple insert concepts generated.
- Initial prototype used to quickly verify fit and assembly.
- 100% outer walls, 15% infill provided adequate flexibility.
- Attaching the insert to the lid proved impractical, as the screw closure made removal difficult, so a top was printed
- Completed two further iterations to refine the dimensions.
- Final design ensured proper lid closure



Waterproofing and Freezing

Hot glue was used to bond the insert into the bottle and form an airtight/watertight seal.

3D print did show some wicking between the extrusion lines using the RS Ideawork printers

The bambu printed lid remained dry

Frozen in the materials lab



Hot Gluing

Extremely sticky. Practice gluing was performed on a sample that was not used for a test to learn more about the process.

The best method seemed to be letting gravity pull the glue down into the interfaces, rather than using sticks or tools to guide the glue.

Further issues include navigating the threaded section of the bottleneck, and gluing the final cap on the bottle.

No “touch-ups” can be made to the lid seal after it is on. This is a very important step in the sealing process.

Water must be added prior to lid gluing.



Gluing Effectiveness

The seal is airtight and watertight.

Several intermittent testing stages were performed during the gluing process - check and re-apply glue to leaking spots.

A freeze-thaw cycle was performed before testing - the structural integrity of the inserts and seal remained sound.

Updates Hot Glue Expenditure: ~£5



Hot Glue and Insert Removal

Removal of the insert and hot glue seal in the bottle can be done with minimal damage to the bottle.

A check was performed prior to insert addition to assess hot glue adhesion properties to the bottle.

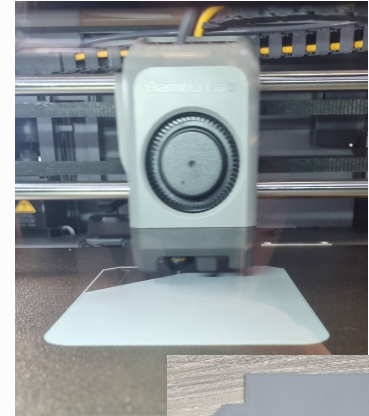
Hot glue delaminates from the HDPE bottle with enough force.

Inserts are damaged in the process, so once they are removed, they cannot be reused.

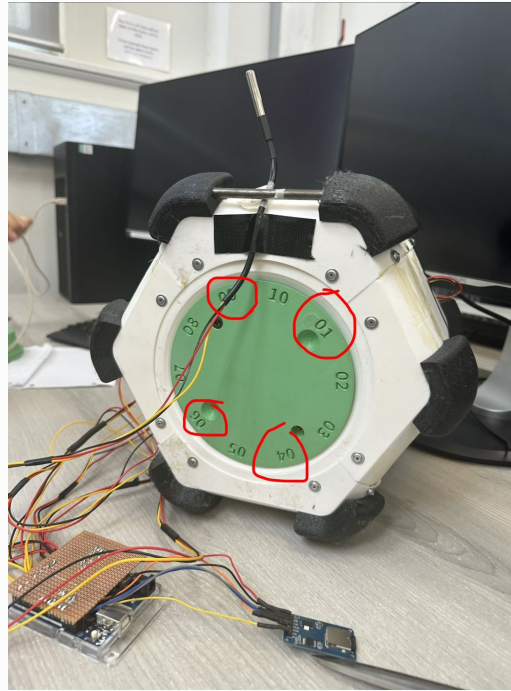
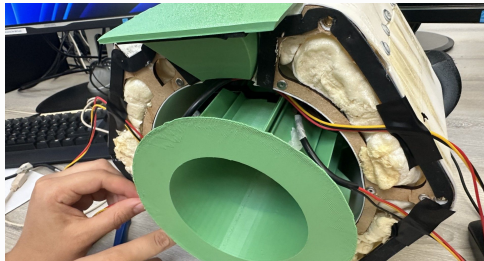
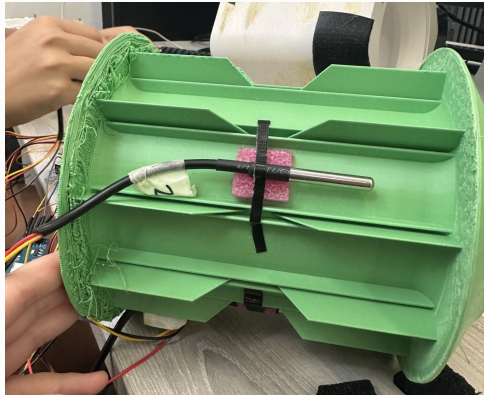


Material Research for Insert Design

- Prototype Material Selection
 - PLA chosen for rapid, low-cost manufacture
 - Hot glue used to create temporary airtight seals for testing.
- Key Issues Identified
 - Water permeation through the printed structure due to print porosity.
 - PLA experiences hydrolytic degradation over time
 - Potential degradation of PLA and adhesive seals during freeze-thaw cycling.
- Design Improvements
 - Higher-quality Bambu printing reduced porosity and improved consistency.
 - PETG identified as a more suitable material due to improved water resistance and takes up less moisture
- Future Manufacturing
 - Current solution suitable for prototyping only.
 - Integrated blow-moulded chambers.
 - Silicon sealant remains flexible, epoxy coating

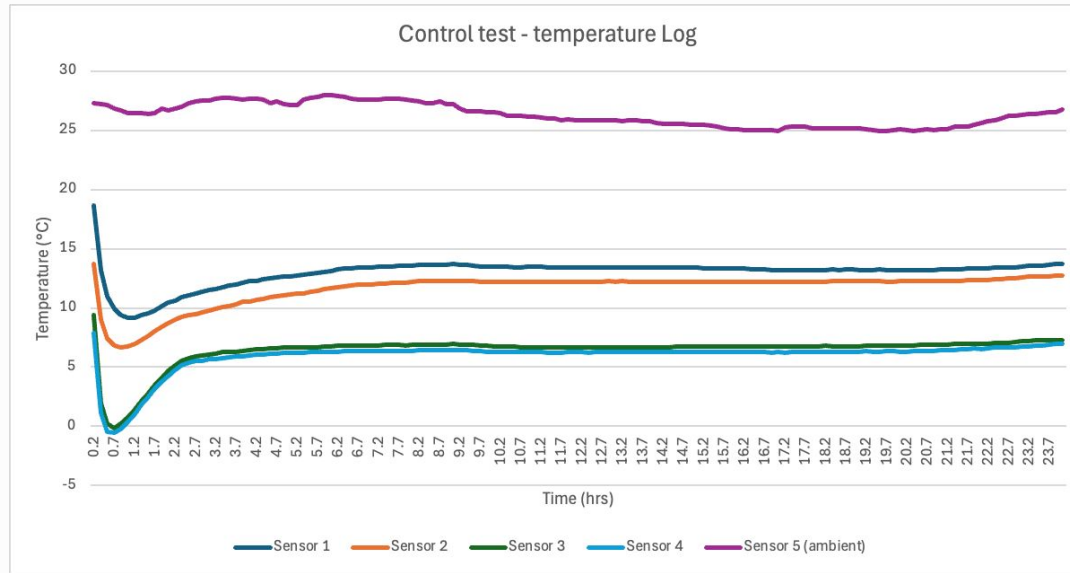


Experimental set-up



Experimental testing

01 Control test - 24 hour control test with the current set-up and the door on top (no inserts).



Results: Initial difference in temperature (1st measurement 5mins in). Temperature drops significantly then rises back to a stable temperature after 5-6 hours

02 Door test - The cooler was oriented upside down with the door on the bottom to test its effects.



Why? Short test, fit into schedule nicely, and also we sought to control this variable if it was a problem

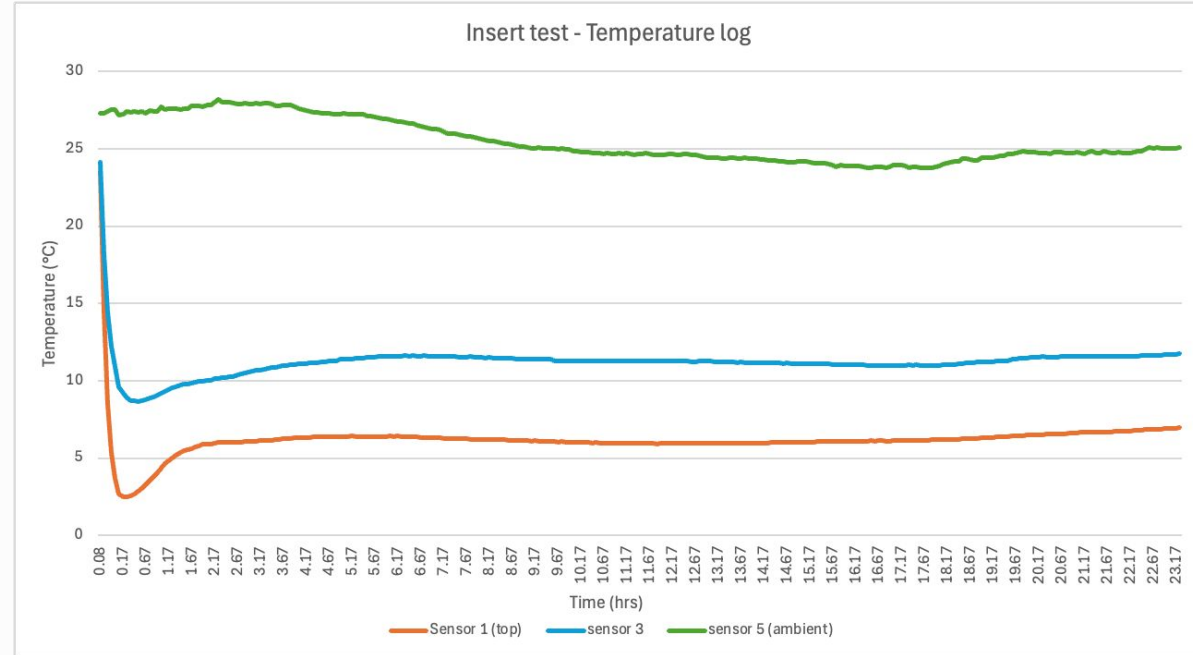
Results: Results remained very similar

03 Insert test - 3D printed inserts were hot glued into the bottle before placing it into the carousel.



At 12 hours:

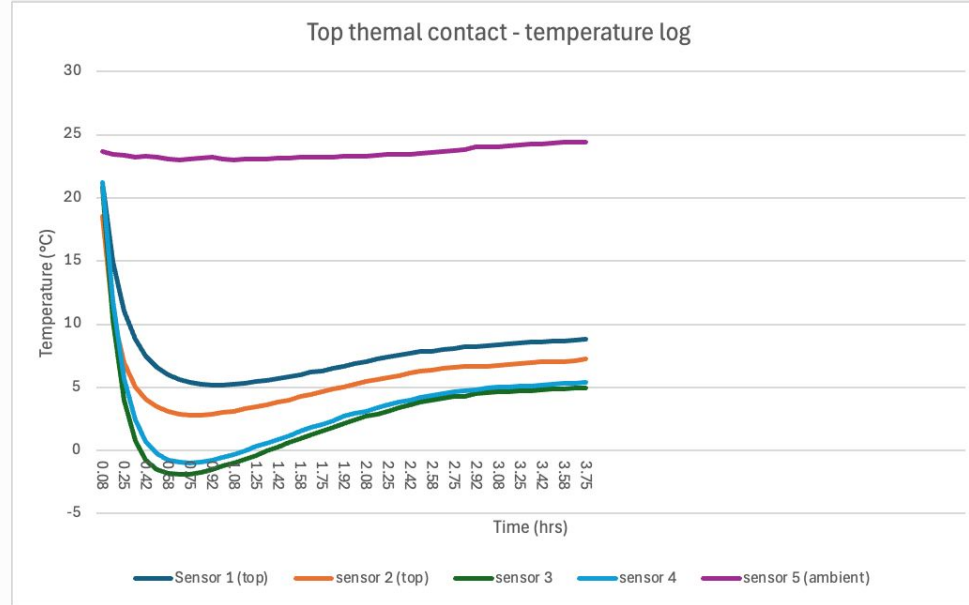
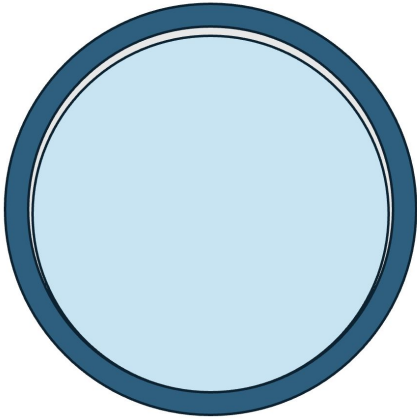
- Insert test:
 $11.19 - 5.94 = 5.25$
- Control test:
 $13.44 - 6.25 = 7.19$



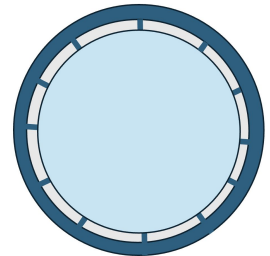
Results: 2 degree improvement to the temperature difference once stabilised

04 Top thermal contact test -

Small inserts (spaghetti) was attached to bottom of central chamber such that ice pack is in contact with top and slightly raised from bottom.

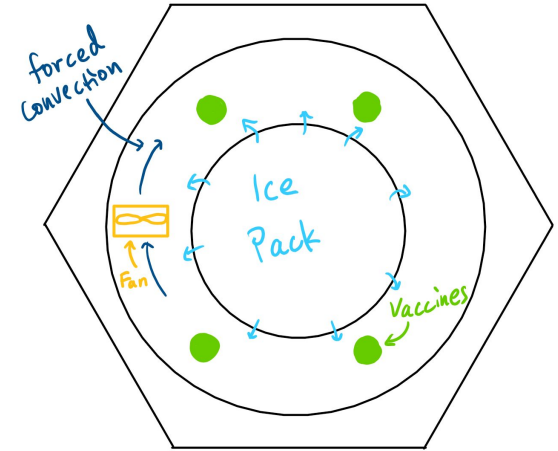
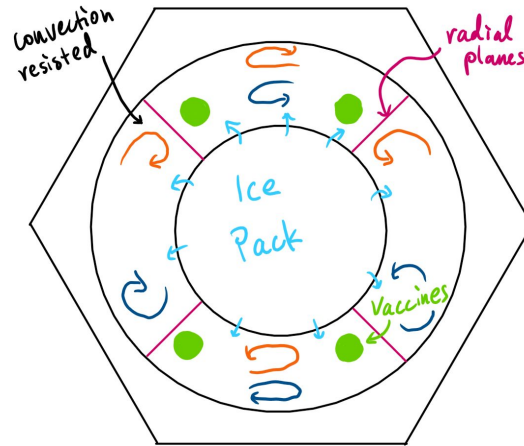
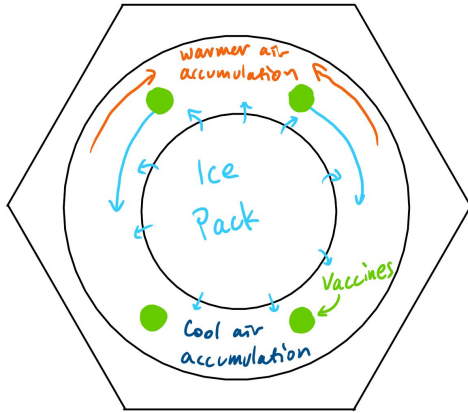


Results:



Convection Research

- Research into convection-based experimental methods and setup designs
 - Convection Resistance Design - radial planes in vaccine carousel
 - Forced Convection - fan-induced convection and mixing



Convection Prototype

A brief model of the experimental setup for the convection resisting design was constructed.

- Assessment of how difficult it may be to get a decent seal within the vaccine chamber.
- Conclusion: very difficult, especially with assembly considerations.

However, it is still worth performing the test to assess the effectiveness of a simple carousel design which may end up mitigating the non-uniform cooling issue.

We want to see, if convection is the issue, if a simple redesign of the carousel can effectively even out the cooling.



Documentation

- Document of images taken throughout the project - important to keep track of specific design choices and for use in presentations or report.
- Meeting summaries consistently taken to keep track of ideas and possible changes to the project.



Meeting 1 18/5



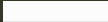
Meeting 2 18/5



Meeting 3 19/5



Meeting 4 20/5



02

Further Issues



Issue Encountered	Mitigation
Delays in sensor procurement affected the original testing schedule	Adapted the schedule and prioritised alternative tasks
Only one cooler bottle was available for testing and the current bottle is sold out	Carefully planned testing schedules to maximise bottle usage
Communication over text messages was sometimes ineffective in conveying ideas	Consistent and increased the frequency of in-person meetings
Early-stage challenges with file organisation and access across the team	Adopted Google Drive for centralised file sharing
Uncertain if the test will fail overnight or the computer has to be turned off	SD card stores any data till that point. We have contingency plans in case we can't access the computer



03

Personal and Technical development

Kerry

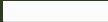
- Attention to detail - The hot gluing process requires patience and a willingness to redo sections to get a sufficient seal.
- Importance of iteration and practice
- Communication with technicians and experts

Kavita

- Code and arduino debugging (COM port issues)
- Soldering skills
- Time management & adaptability

Karen

- Learning practical constraints of 3D printing (porosity, print settings, material selection)
- Iterative designing
- Time management

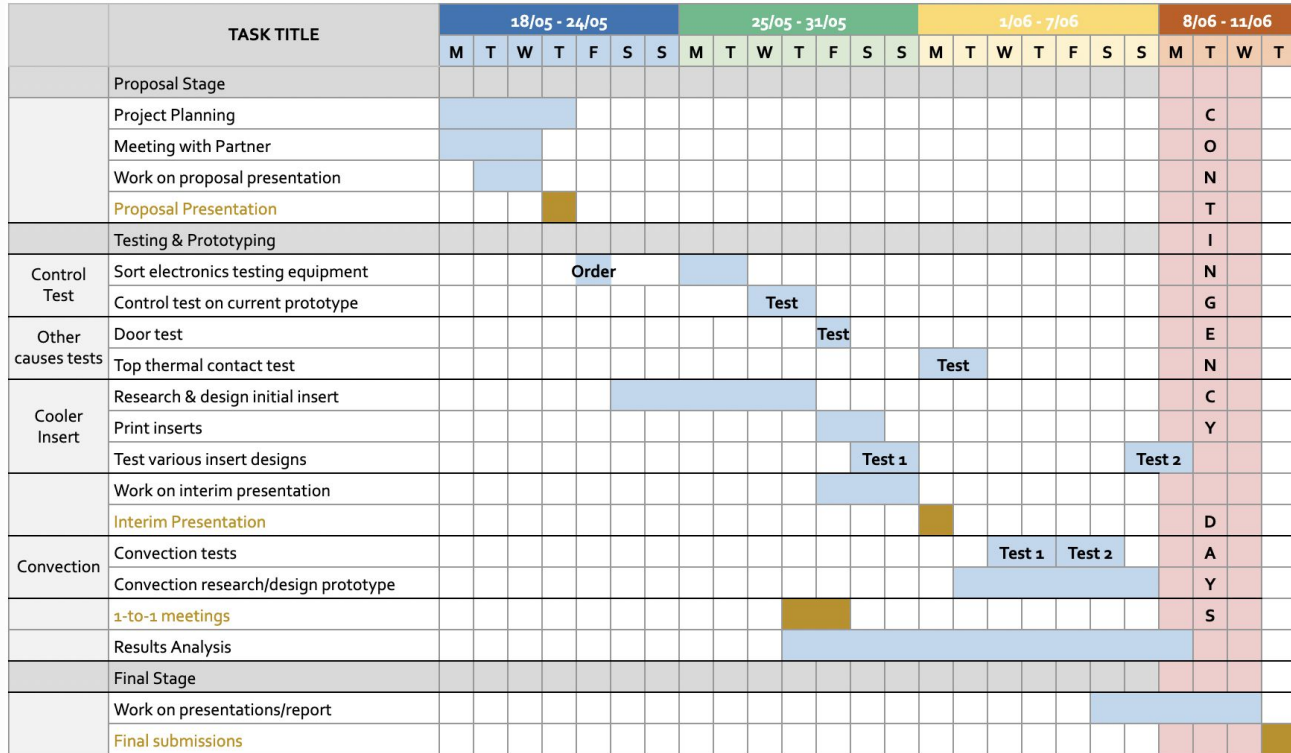


04

Project timeline development



Updated Gantt Chart



Future plans

Key Findings

- There is a large initial temperature difference occurs during the first 10 minutes.
- Door leakage largely ruled out
- Ice segmentation showed some promise of improving cooling uniformity.
- No definitive root cause identified. Further investigation required.
- Better Insulation required in the SMILE Go to keep vaccines in correct range

Further Testing

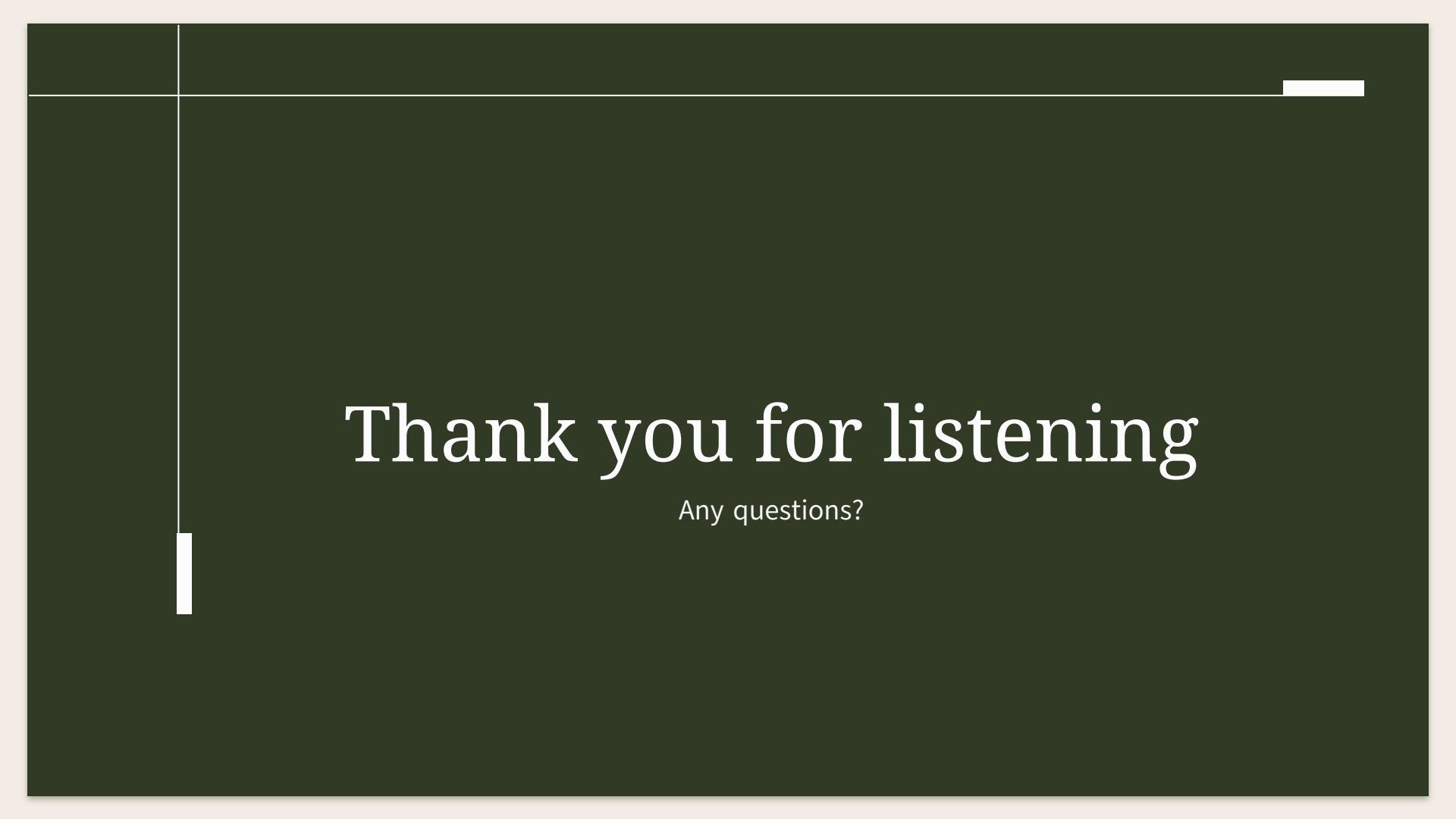
- Convection test - fan (contacted Kevin for purchasing), radial planes.
- Investigate effect of contact of the ice pack with only the lower section of the carousel on non-uniform cooling.

Analyse Results

- Compile the test data to analyse any significant changes.

Design Changes

- Provide possible solutions to any identified causes of non-uniform cooling.



Thank you for listening

Any questions?